Week 3: Quantum Tunneling

Tunelling in Electrical Engineering, Biophysics, and the World

From the Quantum Universe to Quantum Technologies

Week 3 • Spring 2025

1. Understanding Quantum Tunnelling

1.1 Visualizing Tunnelling

When we talk about tunneling, the particle doesn't tunnel through a barrier per se. It isn't even a particle while traveling, it's a wave function with a probability distribution along that wave. That distribution tells us how likely the particle is to be found at any point of the wave once we measure it. Measuring the particle collapses the wave function and makes the particle behave like a particle and not a wave.

In this state, the particle cannot travel through the barrier. However, there is a portion of the wave function that exists on the other side of the barrier. It is much less likely that we measure the particle as being there, but it isn't impossible to do so.

So sometimes when the wave function collapses upon measurement, the particle will have appeared to have "teleported"to the other side of a barrier.

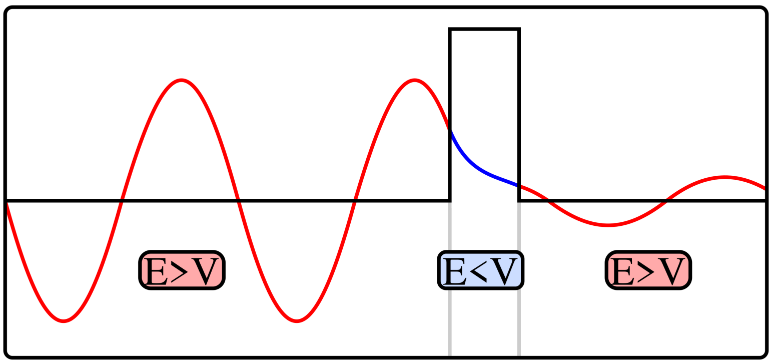

Figure 1: Tunnelling Visualized

1.2 Tunnelling Applications in Biology

One key example in biology is proton tunneling. Spontaneous mutation occurs when normal DNA replication takes place after a particularly significant proton has tunnelled. A hydrogen bond joins DNA base pairs. A double well potential along a hydrogen bond separates a potential energy barrier. It is believed that the double well potential is asymmetric, with one well deeper than the other such that the proton normally rests in the deeper well. For a mutation to occur, the proton must have tunnelled into the shallower well. The proton's movement from its regular position is called a tautomeric transition. If DNA replication takes place in this state, the base pairing might be changed causing mutations.

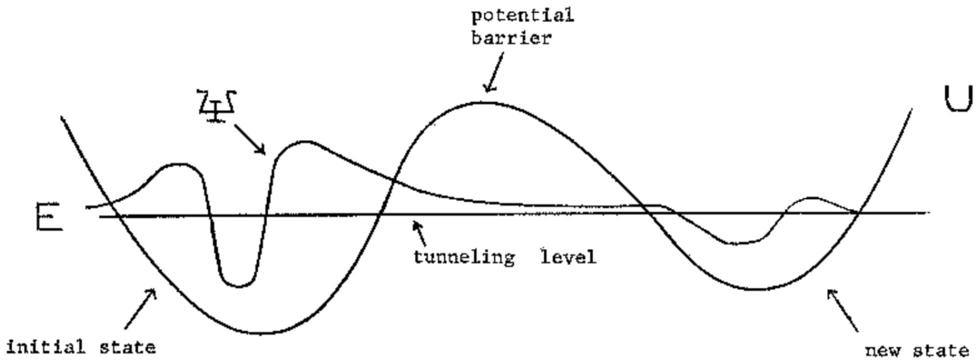

Figure 2: Proton Potential Wells

1. If quantum tunneling did not occur in DNA, how might this affect the rate of genetic mutations and evolution?

2. Besides DNA mutations, can you think of another biological process where quantum tunneling might play a role? Explain your reasoning.

1.3 Tunnelling and Memory in your Computer

1. Flash Memory Basics

USB sticks typically use NAND flash memory, which stores data in memory cells that consist of floating gate transistors. Each cell can hold a charge that represents binary data (0s and 1s).

2. Storing Data

To program a cell (i.e., write data), a voltage is applied to the control gate of the transistor, causing electrons to tunnel through an insulating layer and get trapped in the floating gate. This process is where quantum tunneling comes into play.

3. The Quantum Application

Quantum tunneling is a phenomenon where particles, such as electrons, can pass through a barrier that they classically shouldn't be able to cross due to insufficient energy. In the case of flash memory, this occurs at the nanoscale, where the insulating layer (typically made of materials like silicon dioxide) is thin enough that electrons can tunnel through it when subjected to an electric field.

4. Erasing Data

To erase data, a reverse voltage is applied, allowing the trapped electrons to tunnel back out of the floating gate, returning it to its original state (representing a ’0’). The process is instant, and since electrons stay trapped without power, flash memory doesn’t need a battery to keep your files saved!

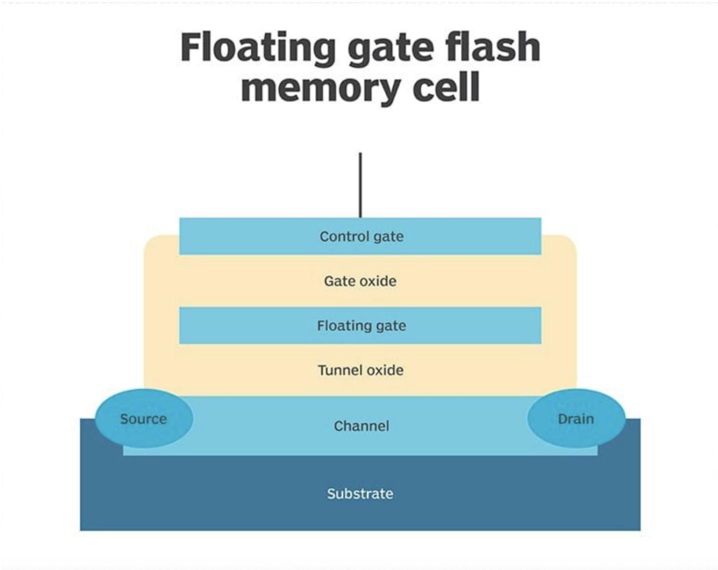

Figure 3: FLASH Memory

1.

3. How does the thickness of the oxide barrier affect the efficiency of quantum tunneling in flash memory?

4. Flash memory has a limited lifespan because the insulating oxide layer degrades over time. How do you think engineers could improve flash memory technology to make it last longer?

5. Why do you think flash memory is more durable and energy-efficient compared to traditional hard drives?